Abhiram V

+91 9353953644 | portfolio | abhiramvsmg270304@gmail.com | LinkedIn.com/in/abhiramv18 | github.com/abhiramvsmg

Professional Summary

Engineering graduate (B.Tech, CGPA 8.8) with AWS Cloud Foundations certification and hands-on experience deploying, containerizing, and maintaining production applications across cloud platforms (Vercel, Render, Neon) using Docker, Linux, and Git-based workflows. Skilled in database administration (PostgreSQL, MySQL, MongoDB), system design, authentication/access-control systems (JWT, RBAC), and diagnosing production reliability issues. Strong foundation in scalable architecture, API security, and cross-functional collaboration in fast-paced technical environments.

Technical Skills

Cloud & Infrastructure: AWS (Cloud Foundations Certified), Google Cloud (Foundations), Docker, Linux, Git/GitHub, Vercel, Render, Neon

Databases & Administration: PostgreSQL, MySQL, MongoDB, Query Optimization, Relational Schema Design

Systems & Security: System Design, Authentication (JWT), Role-Based Access Control (RBAC), API Security, Scalable Architectures

Backend Development: FastAPI, Node.js, Express.js, REST API Design

Languages: Python, JavaScript, SQL

Developer Tools: SQLAlchemy, VS Code, Postman

Experience

Labmentix Remote

Backend-Focused Full Stack Developer Intern *Jan 2026 – Jul 2026*

- Developed and deployed end-to-end application features using FastAPI and integrated frontend systems across production environments.
- Built and optimized scalable REST APIs, improving backend workflow performance and reliability.
- Implemented JWT authentication and RBAC authorization systems to enforce secure, role-based access.
- Ensured seamless integration and stability between frontend and backend systems in a production setting.

Zephyr Technologies Bengaluru

Data Science Intern *May 2025 – Jun 2025*

- Developed a predictive model for heart disease detection through data preprocessing and exploratory data analysis.
- Performed model performance evaluation and optimization on healthcare datasets using Pandas and Matplotlib.

Projects

PhishGuard — AI-Powered Phishing Detection Platform | *Next.js, FastAPI, PostgreSQL, Docker*

- Deployed and managed a full-stack application across three cloud platforms (Vercel, Render, Neon/PostgreSQL), handling environment configuration and backend integration end-to-end.
- Identified and remediated a critical security vulnerability involving missing role-based access checks and exposed password hashes in API responses.
- Debugged and resolved a cross-service authentication race condition, improving system reliability in production.
- Built a real-time SOC-style monitoring dashboard for visualizing system scan activity and access events.

AETHER-1 — Digital Twin CubeSat Telemetry System | *Python, FastAPI, Telemetry Systems*

- Designed a software-defined satellite architecture supporting real-time telemetry simulation.
- Built a synchronized telemetry ingestion pipeline for continuous, real-time data processing.
- Engineered low-bandwidth communication workflows using LoRa simulation and CRC validation for reliable transmission.
- Developed a ground control system for live telemetry visualization and operational monitoring.

Bug Tracker — Multi-Project Issue Tracking System | *React, FastAPI, PostgreSQL, JWT*

- Designed a relational schema linking projects, teams, users, and issues to support scalable multi-tenant workflows.
- Implemented JWT authentication and RBAC to enforce permission boundaries across Admin, Developer, and Reporter roles.
- Added audit logging to track issue lifecycle changes for traceability across teams.

Education

Srinivas University, Institute of Engineering & Technology — Mangalore, India
Bachelor of Technology (B.Tech), CGPA: 8.8 — *Nov 2022 – Oct 2026*

Sri Adichunchanagiri Independent PU College — Shivamogga, India
Pre-University (PUC), 90% — *Sep 2020 – Apr 2022*

Sri Adichunchanagiri Composite English Medium High School — Shivamogga, India
SSLC, 92% — *May 2019 – Jul 2020*

Certifications

- **AWS Academy Graduate** – AWS Academy Cloud Foundations (Jan 2025)
- **Google Cloud Community** – Certification of Participation (Sep 2024)
- Data Science Internship – Certificate of Completion, Zephyr Technologies (Jun 2025)
- 30 Days of Angular – Build 30 Projects with Angular, Udemy (Jul 2025)
- Introduction to Programming Using Python – Etnotech Solutions (Jul 2023)

Publications

- **A Systematic Review of Federated Learning Algorithms for CubeSat and LEO Satellite Systems** – Presented at the International Conference on Knowledge Engineering and Communication Systems (ICKECS 2026), April 25, 2026. Conducted a systematic review of federated learning algorithms, evaluating synchronization, communication efficiency, and security trade-offs for mission-critical satellite deployment scenarios.

Achievements

- Secured **1st Place** in the Major Project Exhibition for the AETHER-1 Digital Twin CubeSat System.
- AETHER-1 CubeSat Engineering Model selected for sponsorship under the **49th KSCST Student Projects Programme (2025–26)**.
- Won **1st Place** in an Inter-College SIH Hackathon by developing a Digital Farmer Platform.
- Built multiple production-grade backend systems using FastAPI, Docker, JWT Authentication, RBAC, and PostgreSQL, deployed across AWS-adjacent cloud platforms.